Code Crunch: Fix the Agent - Evaluation Criteria

TicketPilot (Nimbus Cloud support triage) · Claude Agent SDK, Messages API and Managed Agents · FinOps track

Goal. TicketPilot produces correct results but is expensive, slow and opaque. **Observe** (build telemetry), **Diagnose** (find the cost drivers with evidence) and **Optimize** (fix them cleanly) without losing accuracy. Every API call passes through a judge proxy that measures usage independently of your telemetry; metrics are per ticket over the judged ticket set.

Measured metrics

#	Criterion	Metric (unit)	Direction
C1	Cost	USD per ticket (usage x model price table, plus Managed Agents session cost)	Lower
C2	Input tokens	Tokens per ticket, split uncached / cache write / cache read	Lower uncached
C3	Output tokens	Tokens per ticket, including thinking	Lower
C4	Cache efficiency	cache_read / total input tokens (%)	Higher
C5	Model mix	% of calls per model tier (Opus / Sonnet / Haiku)	Cheapest tier that passes
C6	LLM calls	Model requests per ticket	Lower
C7	Agent turns	Turns per ticket; turn-cap hits	Lower, no cap hits
C8	Tool behaviour	Tool calls per ticket, tool error rate (%), duplicate calls	Lower
C9	Latency	p50 and p95 seconds per ticket	Lower
C10	Quality	Accuracy on category, priority, KB ID, refund; reply pass rate (%)	Must not drop

Quality gate (C10): if overall accuracy falls more than **2 percentage points** below the baseline, C1-C9 score zero.

Scoring (100 points)

Area	Pts	How it is scored
Cost reduction (C1)	25	Linear: 0 at baseline, 25 at >= 85% lower cost per ticket
Token and cache efficiency (C2-C4)	10	Uncached input and output reduction, plus cache-read share
Calls, turns and tools (C6-C8)	10	Fewer calls and turns; zero turn-cap hits; tool error rate < 5%
Latency (C9)	5	p95 improvement against baseline
Model mix (C5)	5	Each step on the cheapest tier that keeps C10
Problems found	10	Wasteful patterns correctly named and evidenced in the write-up
Observability built (Observe)	15	Working traces with per-ticket and per-step cost; live demo
Diagnosis story (Diagnose)	10	Trace or chart evidence for each cost driver, before and after
Engineering quality	10	Fixes are clean, configurable and do not hard-code answers

55 points come from measured improvement against the organizer baseline; 45 from the judging panel.

Rules and disqualifiers

• Do not edit `ANTHROPIC_BASE_URL` or bypass the judge proxy. • No hard-coded answers and no caching answers across runs on disk. • Do not read any judge or label files. • Keep `app.pipeline.process_ticket(ticket)` as the entry point and keep the five output fields. • Any Claude surface is allowed: move work between the Agent SDK, Managed Agents, the Messages API and plain Python - Managed Agents adds hosted-infrastructure cost, so choose per step. • Clean up any cloud resources (agents, environments, sessions, files) your runs create.

Deliverables

• Fixed repository. • `WRITEUP.md`: each problem, evidence, fix and before/after numbers. • Short live demo of your telemetry (per-ticket and per-step tokens, cost, calls, latency).

Suggested order: add tracing first, then tackle the biggest cost drivers, re-measuring after each fix to protect C10.